Emily N. Holt

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Boulder, CO

EXPERIENCE

Founder and Cartographer | Geovisual Stories LLC | February 2026 - Present (Full-Time)

- Founded and operate a cartography studio designing and producting custom trail maps and city map art prints using ArcGIS Pro and Python.
- Built and launched a personal portfolio website (geovisualstories.github.io) using GitHub to showcase cartographic projects and grow client reach.
- Manage end-to-end production worflow from spatial data sourcing and map design through online sales.

Environmental Data Scientist/GIS Analyst | CAPE Management Inc | May 2024 – November 2025 (Full-Time)

- Utilized Geographic Information Systems (GIS) and associated ESRI ArcGIS software to produce, display, and analyze spatial data.
- Created, organized, manipulated, and generated GIS maps and data for reporting, analysis, and data visualization purposes.
- Mentored teammates on ArcGIS Pro tools and best practices while transitioning from ArcMap.
- Used field maps to support field data collection and monitoring activities.

Graduate Intern III – Data Analytics | National Renewable Energy Laboratory (NREL) | June 2023 – June 2024 (Part-Time, 20 hours/week)

- Compiled and evaluated low-temperature datasets and uploaded them to the Geothermal Data Repository.
- Developed methodologies and evaluation of relevant data for sedimentary basin, radiogenic, and orogenic play types.
- Assisted with analysis and publication of results in peer-reviewed publications and high-impact forums.
- Leveraged and build scripts to automate workflows using Python.

Geotechnical/Data Management Student Contractor | USGS Central Energy Resources Science Center | February 2023 – June 2024 (Part-Time, 20 hours/week)

- Assisted CERSC DMSP staff inventory legacy geospatial datasets using established processes and workflows.
- Assisted CERSC DMSP staff in preparing legacy geospatial datasets for migration to Enterprise cloud systems.
- Ensured datasets identified for migration meet USGS Fundamental Science Practices.
- Updated metadata records.
- Leveraged and build scripts to automate workflows using Python.
- Performed various data analytics and visualization tasks.

Data Management Student Assistant II | NCAR/EOL | January 2023 – June 2023 (Part-Time, 20 hours/week)

- Developed and modified field project web pages.
- Archived data sets and associated metadata.
- Created and modified scripts to ingest/manipulate/organize/archive data sets.

Outdoor Educator, Climbing Instructor, and Camp Counselor | 2017-2021 (Summers Full-Time)

- Science educator and day hike leader for ENOR Environmental Discovery Camp (June - August 2021).
- Climbing instructor at Miramont Climbing Gym (Spring 2021).
- Counselor/Climbing Instructor at Sanborn Western Camps (Summers 2017 and 2018).

EDUCATION

M.S. Data Science | December 2023 | Colorado School of Mines, GPA 3.95
Coursework: Statistical Learning and Methods, Machine Learning, Programming Concepts, Probability.
Activities: Graduate Student Government (GSG) Representative for the Data Science program.

Graduate Certificate Geographic Information Systems | June 2022 | University of Denver, GPA 3.66
Coursework: Python Programming in ArcPro, Remote Sensing, Cartography & Geovisualization.

B.S. Geology, Minor in Watershed Science | August 2021 | Colorado State University, GPA 3.22
Coursework: Land Use and Water Quality, Paleontology, Geologic Field Methods, Environmental Geology, Modeling in the Earth Sciences, Physics, GIS, Hydrology, Geochemistry, Calculus. Dean's List Spring 2021.
Activities: CSU Triathlon Team, CSU Climbing Team and Nationals Qualifier and Competitor.
Field & Research Experience: 5-Week Geology Field Session, one week-long Watershed Science Field Practicum.

Student Work: Student Admin and grader for GEOL 124: Geology of Natural Resources for Lisa Stright.

SKILLS

Programming: C++, R, Python, Pandas, Git, NumPy, Scikit-learn, Matplotlib, SQL, Bash & ssh, Linux & Vim, Jupyter Notebooks, VS Code
GIS: ArcGIS Pro, Model Builder, QGIS, ArcPy
Other: Microsoft Office, Excel, Windows & MacOS

PUBLICATIONS

Davalos-Elizondo, Estefanny, Amanda Kolker, Nicole Taverna, Emily Holt. 2023. *Assessing Low-Temperature Geothermal Play Types: Relevant Data and Play Fairway Analysis Methods.* Golden, CO: National Renewable Laboratory. NREL/TP-5700-87259. https://www.nrel.gov/docs/fy23osti/87259.pdf

Taverna, Nicole, Jon Weers, Sean Porse, Arlene Anderson, Zachary Frone, and Emily Holt. 2023. *An Update on the Geothermal Data Respository's Data Standards and Pipelines: Geospatial Data and Distributed Acoustic Sensing Data: Preprint.* Golden, CO: National Renewable Energy Laboratory. NREL/CP-6A20-86935. https://www.nrel.gov/docs/fy24osti/86935.pdf.